

Lab 3 – Architecture Justification

Faculty Research Grant & Publication Tracker

Architectural Choice

We chose a **Layered Architecture** for the Faculty Research Grant & Publication Tracker. The system separates user interaction, business processing, and data management. The component diagram represents these responsibilities through the Faculty Researcher Interface, Grant Management, Publication & Approval Management, Dean Approval & Analytics, and Research Database components.

Reason 1 – Separation of Responsibilities

The architecture separates different responsibilities into dedicated components. The interface handles interaction with Faculty Researchers and the Research Dean. Grant Management handles grant allocation, procurement expenses, and balance validation. Publication & Approval Management handles publication information and approval requests, while Dean Approval & Analytics manages dean decisions and fund analytics. This separation makes the system easier to understand and maintain.

Reason 2 – Maintainability

Changes can be made within one functional area with limited impact on other parts of the system. For example, changes to publication or co-author approval processing can be implemented in the Publication & Approval Management component without changing the user interface or research database structure. This modular organization supports easier testing, debugging, and future enhancement.

Security Advantage

The layered structure allows validation and authorization checks to be performed in the business layer before financial or approval operations reach the database. Grant expenses can be validated against the available balance, and approval actions can be restricted to authorized users such as the Research Dean. The Audit Ledger provides tamper-evident records of financial and approval-related modifications, improving traceability.

Performance Benefit

Separating business processing from data management allows grant calculations and analytics to be handled efficiently without mixing presentation logic with database operations. The Research Database provides centralized access to grant, publication, approval, and analytics data. This organization supports the requirement that fund burn-up analytics should be generated within the specified response time for faculty grant records.

Conclusion

Overall, Layered Architecture provides a clear structure for the Faculty Research Grant & Publication Tracker, with separate responsibilities, easier maintenance, controlled access to sensitive operations, and organized data processing.